

PRACTICAL 1 &2

Q.1)

Part 1: C Program using fork() and exec()

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char input[100];
    char *args[10];

    printf("Enter a Linux command (e.g., ls or date): ");
    fgets(input, 100, stdin);
    input[strcspn(input, "\n")] = 0;
    int i = 0;
    args[i] = strtok(input, " ");
    while (args[i] != NULL) {
        i++;
        args[i] = strtok(NULL, " ");
    }

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    } else if (pid == 0) {
        printf("[Child] My PID: %d, Parent PID: %d\n", getpid(), getppid());
        execvp(args[0], args);
        printf("Command execution failed! Make sure you typed it right.\n");
        exit(1);
    } else {
        printf("[Parent] My PID: %d. Waiting for child (PID: %d) to finish...\n", getpid(), pid);
        wait(NULL);
        printf("[Parent] Child finished executing. Exiting.\n");
    }

    return 0;
}
```

```
praks@praks:~$ nano task1.c
praks@praks:~$ gcc task1.c -o task1
./task1
Enter a Linux command (e.g., ls or date): date
[Parent] My PID: 755. Waiting for child (PID: 756) to finish...
[Child] My PID: 756, Parent PID: 755
Fri Aug 7 09:22:50 UTC 2026
[Parent] Child finished executing. Exiting.
```

Part 2: Hardware Resources Investigation

1. uname -a

```
praks@praks:~$ uname -a
Linux praks 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux
```

2. lscpu

```
praks@praks:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                  6
Model:                      186
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.00
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cxi6 pcid sse4_1 sse4_2 x2apic movbe p
opcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 eras invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vmmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_lid arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                         384 KiB (8 instances)
L1i:                         256 KiB (8 instances)
L2:                           10 MiB (8 instances)
L3:                           24 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):           0-15
Vulnerabilities:
Gather data sampling:         Not affected
Indirect target selection:    Not affected
Itlb multihit:               Not affected
L1tf:                         Not affected
Mds:                          Not affected
Meltdown:                    Not affected
Mmio stale data:              Not affected
Reg file data sampling:       Mitigation; Clear Register File
Retbleed:                    Mitigation; Enhanced IBRS
Spec rstack overflow:         Not affected
Spec store bypass:            Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                   Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                   Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI BHI_OIS_S
Srbds:                        Not affected
Tsx:                          Not affected
Tsx async abort:              Not affected
Vmscape:                      Not affected
```

3. lsblk

```
praks@praks:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 364.8M  1 disk
sdb   8:16   0 144.5M  1 disk
sdc   8:32   0     2G   0 disk [SWAP]
sdd   8:48   0     1T   0 disk /mnt/wslg/distro
```

4. ps -ef | head -n 10

```
praks@praks:~$ ps -ef | head -n 10
UID        PID     PPID  C  STATE TTY      TIME   CMD
root         1         0  0   0:05 ?        00:00:01 /sbin/init
root         2         1  0   0:05 ?        00:00:00 /init
root         6         2  0   0:05 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        49         1  0   0:05 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+    86         1  0   0:05 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root        92         1  0   0:05 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root       126         1  0   0:05 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root       131         1  0   0:05 ?        00:00:00 /usr/sbin/cron -f -P
message+   138         1  0   0:05 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
```

5. top -b -n 1 | head -n 15

```
praks@praks:~$ top -b -n 1 | head -n 15
top - 09:30:39 up 36 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.6 sy, 0.0 ni, 98.9 id, 0.6 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7758.1 total, 7145.2 free, 545.1 used, 219.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7212.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0  24524  15220 11352 S   0.0   0.2   0:01.75 systemd
    2 root        20   0   3172   2204   2072 S   0.0   0.0   0:00.04 init-sy+
    6 root        20   0   3172   2044   1940 S   0.0   0.0   0:00.00 init
   49 root        19  -1  50380  16576 15296 S   0.0   0.2   0:00.43 systemd+
   86 systemd+    20   0  22400  14164 11672 S   0.0   0.2   0:00.16 systemd+
   92 root        20   0  35252  12200  9072 S   0.0   0.2   0:00.83 systemd+
  126 root        20   0   2880   1916   1792 S   0.0   0.0   0:00.09 chronyd+
  131 root        20   0   4464   3080   2832 S   0.0   0.0   0:00.02 cron
```

Q.2)

```
praks@praks: ~  
GNU nano 8.7.1 b.c  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main()  
{  
    int src, dest;  
    char buffer[1024];  
    int bytes;  
  
    src = open("input.txt", O_RDONLY);  
  
    if(src < 0)  
    {  
        printf("Cannot open source file\n");  
        return 1;  
    }  
  
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)  
    {  
        write(dest, buffer, bytes);  
    }  
  
    close(src);  
    close(dest);  
  
    printf("File copied successfully.\n");  
  
    return 0;  
}
```

```
praks@praks:~$ nano b.c  
praks@praks:~$ gcc b.c - b  
gcc: error: -E or -x required when input is from standard input  
praks@praks:~$ gcc b.c -o b  
praks@praks:~$ echo "Hello, Ubuntu!" > input.txt  
praks@praks:~$ ./b  
File copied successfully.
```

HISTORY

```
66 nano task1.c
67 gcc task1.c -o task1
68 ./task1
69 uname -a
70 lscpu
71 lsblk
72 ps -ef | head -n 10
73 top -b -n 1 | head -n 15
74 echo "Hello OS Lab" > sample.txt
75 strace cat sample.txt
76 sudo apt install strace
77 nano b.c
78 gcc b.c - b
79 gcc b.c -o b
80 echo "Hello, Ubuntu!" > input.txt
81 ./b
82 nano b.c
83 history
praks@praks:~$ |
```